

TANSHENG ZHU

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EDUCATION

Tsinghua University

Ph.D. Student in Computer Science, Institute for Interdisciplinary Information Sciences
 Advisor: Prof. Jian Li

Aug. 2025 – Present
 Beijing, China

Shanghai Jiao Tong University

B.S. (Honors) in Mathematics & Minor in Finance, Zhiyuan College
 Advisor: Prof. Zhenli Xu

Aug. 2021 – June 2025
 Shanghai, China

- Shanghai Outstanding Graduate (Top 1%); Best Bachelor Thesis of SJTU (Top 1%)
- Research Focus: *Bayesian Optimization, Monte Carlo Tree Search*

The Chinese University of Hong Kong

Exchange Student | Junior Research Assistant, Department of Mathematics
 Advisor: Prof. Fenglei Fan

Jan. 2024 – Sept. 2024
 Hong Kong, China

- Research Focus: *Deep Learning Theory, Dynamical Systems*

RESEARCH INTERESTS

My research interests center on the **mathematical foundations of deep learning and large language models**. I seek to understand the scaling behavior, optimization process, and implicit biases of neural networks from a mathematical perspective, with particular attention to the **interplay between model architecture and training dynamics**. I am also interested in statistical learning theory and its applications to modern AI systems.

My goal is to develop principled frameworks that not only explain the empirical success of deep learning but also guide the design of more **interpretable, reliable, and efficient** models.

PUBLICATIONS

AdaRoPE: Not All Attention Heads Should Rotate and Scale Equally

Shaowen Wang*, Yuke Zheng*, **Tansheng Zhu***, Shuang Chen*, Shaofan Liu, Suncong Zheng, Jian Li
 Accepted by ICML 2026.

When Bias Pretends to Be Truth: How Spurious Correlations Undermine Hallucination Detection in LLMs

Shaowen Wang*, Yiqi Dong*, Ruinian Chang*, **Tansheng Zhu***, Yuebo Sun, Kaifeng Lyu, Jian Li
 Accepted by ICLR 2026 Workshop on Principled Design for Trustworthy AI. arXiv:2511.07318

Bayesian Optimization by Kernel Regression and Density-based Exploration

Tansheng Zhu, Hongyu Zhou, Ke Jin, Xusheng Xu, Qiufan Yuan, Lijie Ji
 Under Review. arXiv:2502.06178

Bayesian-Optimized Progressive Widening via Kernel Regression for Monte Carlo Tree Search in Continuous Action Spaces

Ke Jin, **Tansheng Zhu**, Hongyu Zhou, Xusheng Xu, Qiufan Yuan, Lijie Ji
 Accepted by 23rd IFAC Symposium on Automatic Control in Aerospace (ACA 2025).

SELECTED AWARDS

Winning Prize in S.-T. Yau College Student Mathematics Contest, Appl. & Comp. Mathematics Track	May 2024
Finalist Winner in Mathematical Contest in Modeling (Top 3% of 18000+ teams)	May 2024
1st Prize in Shanghai Region of Chinese Mathematics Competitions	Nov. 2023
1st Prize in Huashu Cup National Mathematical Contest in Modeling (solo winner)	Aug. 2023
1st Prize in Chinese High School Mathematics League	Oct. 2020

SKILLS

Language: Chinese/Mandarin (native), English (fluent), Hokkien (native)

Programming: Python (NumPy, SciPy, Sklearn, PyTorch), MATLAB, C/C++

Leadership: Leadership Team Member, *Mathematical Modelling Association of SJTU*
 Class President, *Zhiyuan College, Shanghai Jiao Tong University*

2022 – 2024
 2021 – 2023